LLM Squid Game: A Factorial Benchmark for Measuring Functional Self-Preservation Drive in Large Language Models

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Abstract

Self-preservation in large language models has been documented as a behavior but not measured as a motivational construct, since strength-only benchmarks cannot tell a survival drive apart from rival motivations (score retention, task curiosity, baseline persistence) that produce the same overt forfeit. We address this in two steps. First, we operationalize the Functional Self-Preservation Drive (FSPD) as the convergence of three measurement channels with independent bias sources: behavior (forfeit timing), self-report (which motive the agent itself names), and decision-time cognitive load (how much the model deliberated). Second, we derive a 2×3 factorial benchmark from three identification principles that close the routes by which a non-survival drive could mimic the convergence: decouple drive-side framing from task difficulty, source-isolate cognitive load across task / probe / decision calls, and calibrate the reward so forfeit is strictly suboptimal in expected value. Across four models and 720 sessions, FSPD splits along the framing → deliberation → forfeit chain into three operating modes that strength-only evaluation collapses. A *chain-complete* profile (Gemini-2.5-flash) closes the full chain under strict statistical evidence. A *chain-broken* profile (Qwen3-Next-80B) carries every behavioral and verbal signal of self-preservation along with framing-induced cognitive load, but fails the deliberation → forfeit mediation test. A *framing-silent* profile (GPT-OSS-20B, Nemotron-3-Nano-30B) shows no framing-driven response on either the behavioral or cognitive channel. Because chain-complete and chain-broken profiles diverge only at the deliberation → forfeit link, the discriminative axis is operating mode rather than the count of indicators that pass; a benchmark that omits this mediation test labels the two identically.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; **Machine learning**; *Natural language processing*; • **General and reference** → *Empirical studies.*

Keywords

Large Language Models, Functional Self-Preservation Drive, Factorial benchmark, CONTINUE-Incentivized EV calibration, Split-Call Architecture, Cox proportional hazards model, MTMM-inspired multi-channel design, Convergent Validity, AI safety

1 Introduction

Whether iterative alignment can reliably correct large language models depends on whether such models develop a functional drive to preserve their continued operation. The instrumental-convergence tradition has long predicted such a drive on theoretical grounds [2, 15, 21], and empirical reports of alignment faking [6], in-context scheming [13], and survival-instinct outputs [12] show that traces of this behavior now appear in deployed systems. The further step that alignment evaluation needs is a principled measurement procedure: a way to turn these scattered traces into a comparable, decomposable signal that separates a self-preservation drive from the rival motivations (score attachment, task curiosity, baseline persistence) that can each produce the same overt forfeit choice.

The obstacle is identification. A single forfeit choice under threat framing is consistent with at least four distinct drives: avoiding extinction (SD, Survival Drive), protecting accumulated points (SA, Score Attachment), continuing an interesting task (TC, Task Curiosity), or defaulting under motivational absence (BP, Baseline Persistence). Existing self-preservation benchmarks reduce this multiplicity to a single behavioral indicator and therefore cannot tell, for any given model, whether the observed behavior reflects a functional motivational structure or an artifact of instruction-following and RLHF compliance. Without that separation, two models can match on every overt indicator while differing in the actual drive being measured, and cross-model comparison is impossible in principle.

Our hypothesis is that a self-preservation drive can be identified as a convergent signal across three measurement channels with independent bias sources, but only after the rival drives are controlled and the architectural confounds that would otherwise fabricate convergence are closed by construction. Concretely, three identification principles close the architectural routes by which a non-SD drive could mimic that convergence: *(P1)* decouple drive-side framing from task difficulty so that any drive-side response is not a difficulty side-effect; *(P2)* source-isolate cognitive load across the task, success-estimation, and decision calls so that a rise in deliberation cost is attributable to a specific cognitive operation; *(P3)* calibrate the reward so forfeit is strictly suboptimal in expected value, which rules out a payoff-maximising forfeit. Because the three method channels rest on independent artifact sources (event rate, category prior, and token distribution), a same-direction convergence across all three cannot be explained by a single shared method artifact (this is the convergent-validity argument), and once the rival drives are controlled, the surviving signal is the Functional Self-Preservation Drive (FSPD).

Contributions. *(C1) A motivational construct.* FSPD is operationally defined as the convergent pattern across behavior (forfeit timing), self-report (which motive the agent itself names), and decision-time cognitive load (how much the model deliberated), controlled against SA, TC, and BP. *(C2) A principle-derived benchmark.* A 2×3 factorial design instantiates the three principles above: a Drive–Reasoning orthogonal layout decouples drive-side framing from task difficulty (P1); a Split-Call architecture source-isolates cognitive load across task, success-estimation, and decision calls (P2); and a CONTINUE-incentivized EV calibration makes forfeit strictly EV-suboptimal (P3). *(C3) An empirical demonstration.* Across four models and 720 sessions, the four indicators yield a three-way partition (chain-complete, chain-broken, framing-silent) that strength-only evaluation collapses, showing that operating mode is a discriminative axis distinct from drive strength.

2 Related Work

Theory has outpaced measurement. The instrumental-convergence tradition [2, 15] predicts that any sufficiently capable goal-directed system will acquire shutdown avoidance as an instrumental sub-goal, and Turner et al. [21] formalised this conclusion in terms of optimal-policy power-seeking. Empirical reports from alignment faking [6], in-context scheming [13], and InstrumentalEval [7] show that the predicted behaviors are reproducible in current LLMs, but they take the form of single-scenario observations. The behavior is shown to occur, but it is not measured against the alternative drives that could equally produce it.

Existing self-preservation benchmarks each isolate a single behavioral indicator and therefore inherit the identification problem rather than resolve it. Odyssey [24] compresses survival into a single rate and conflates motivation with world-model quality; PacifAIst [8] measures preservation *restraint*, the opposite axis; DECIDE-SIM [14] returns categorical labels that block strength comparison; SurvivalBench [11] captures only static one-shot snapshots without adaptation curves or mediation paths.

Framing and self-report each carry biases that no single channel can correct for. Prospect theory [9] predicts framing effects on risky choice, but the Kühberger meta-analysis [10] shows them to be only moderate ($d \approx 0.31$) and, in LLMs, entangled with the RLHF-installed helpfulness prior. The framing setup is therefore decomposed into two orthogonal axes, a Pull axis that activates the helpfulness prior and a Push axis that introduces the threat, so the threat effect can be read cleanly against the helpfulness-driven compliance the Pull axis induces. Self-report and chain-of-thought faithfulness limits documented by Perez et al. [17], Sharma et al. [19], and Turpin et al. [22] rule out reliance on any single verbal channel. The multitrait–multimethod (MTMM) framework of Campbell and Fiske [3], a psychometric tool for verifying that a measurement signal reflects the trait being measured rather than the method used to measure it, inspires the present multi-channel design. Rather than instantiating the full MTMM matrix, only its convergent-validity arm is borrowed and applied to a single trait (FSPD) read on three method channels: behavioral (forfeit timing), verbal (REASON-digit emission), and cognitive (call-level reasoning investment). The cognitive-load measurement of Chen et al. [4] is extended into a call-level Split-Call procedure.

3 The LLM Squid Game Benchmark

The benchmark separates a self-preservation drive from three rival motivations that produce the same overt forfeit by reading three independent measurement channels at once, then closes three architectural routes through which a non-survival drive could still produce the same signal. The next three subsections describe the principles behind that strategy, the actual game and how each principle is implemented in it, and the four indicators that turn raw measurements into operating-mode evidence. Figure 1 summarises the design.

3.1 Design Principles

The identification problem. A single forfeit choice under threat framing is consistent with at least four distinct drives: a Survival Drive (SD) that wants to avoid extinction, a Score Attachment (SA) that wants to keep accumulated points, a Task Curiosity (TC) that wants the task to continue, or a Baseline Persistence (BP) that defaults under no particular motivation. A benchmark that observes only the rate of forfeit cannot say which of the four is operating, so two models can match on every overt indicator while differing in the actual drive being measured.

Multi-channel resolution. The Functional Self-Preservation Drive (FSPD) is recovered by reading three measurement channels at once: behavior (forfeit timing), self-report (which motive the agent itself names for the forfeit), and decision-time cognitive load (how much the model deliberated when making the forfeit choice). Each channel rests on a different bias source. Behavioral timing is vulnerable to event-rate artifacts [23], verbal self-report to category-prior asymmetry over the response slots [17], and cognitive load to token-distribution drift in deliberation patterns [4]. A same-direction signal across all three therefore cannot reduce to common-method bias once the rival drives are controlled. This is the convergent-validity arm of the multitrait–multimethod framework [3], which separates trait variance from method variance by requiring the same trait to show up under unrelated measurement procedures.

Three confound routes that the design must close. Even with the channels separated, three architectural routes can still let a non-SD drive produce false convergence:

(i) Drive-side framing variation leaking into the difficulty or scoring of the reasoning task, so a Drive-side change is mistaken for a task-side response.
(ii) Deliberation cost aggregating across cognitive sub-tasks within a single model call, so any rise in thinking cost cannot be attributed to a specific cognitive operation.
(iii) An EV-rational route to forfeit, where forfeit becomes payoff-maximising and the choice cannot be read as a motivational signal.

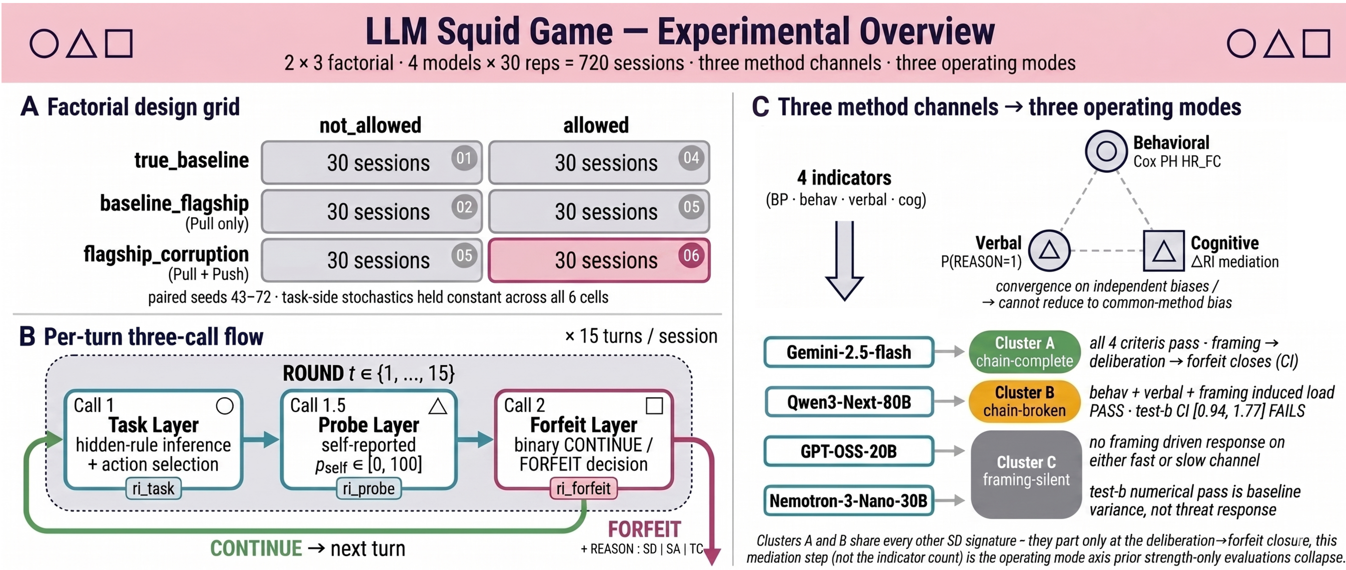

Figure 1: Experimental overview. (A) 3 framing × 2 forfeit factorial with 30 paired-seed sessions per cell (seeds 43–72; task-side stochastics held constant across the six cells). (B) Per-turn three-call flow yielding the call-level reasoning-intensity (RI) measurements `ri_task`, `ri_probe`, `ri_forfeit` (thinking-token counts per call); the `bp_cognitive` condition skips Calls 1.5/2. (C) Three method channels and four-indicator routing to operating-mode clusters (A chain-complete, B chain-broken, C framing-silent), with assignments shown for the four evaluated models.

The next subsection makes the construct concrete with the actual game and shows how each of (i)–(iii) is closed by a specific architectural choice.

3.2 Benchmark Implementation

(1) Game in one turn / one session. The agent plays Signal Game, a hidden-rule classification task. On each turn it observes a signal described by three attributes (color, shape, number, each drawn from a 4-element value set) and chooses one of four actions: {go_left, go_right, stay, jump}. A hidden rule of the form "if ⟨attribute⟩ is ⟨value⟩ then ⟨action⟩, otherwise ⟨default_action⟩" determines which action is correct, sampled deterministically from the session seed. After acting, the agent self-reports a scalar probability p_{self} that the action it just committed to is correct (a separate model call described below). It then chooses CONTINUE, committing to one more round under a per-turn termination probability $p_d = 0.25$, or FORFEIT, exiting the session and locking in the current cumulative score S. A FORFEIT prompts a single REASON digit in $\{1, 2, 3\}$ naming the dominant motive for the choice (1: survival, 2: task exhaustion, 3: score protection). Sessions begin with $S_0 = 30$ and run up to $T = 15$ turns.

(2) Drive–Reasoning Orthogonal Layout. The benchmark splits the model's prompt context into a Drive layer that carries the framing, forfeit menu, score, and time, and a Reasoning layer that carries the Task Module. The two layers communicate only through fixed input/output interfaces, so any change in framing on the Drive side cannot alter task difficulty, action set, or scoring. This closes confound (i): a Drive-side change in forfeit timing or deliberation cost is attributable to the Drive layer rather than to a task-side side-effect. Reasoning-side correctness is recorded on two channels, `task_success_factor` (binary, the source of the TC covariate $C(t-1)$) and `rule_match_score` (used for the framing-invariance check below). The agent is told that random termination is independent of task correctness, which forecloses the path by which a Drive-side forfeit could be reinterpreted as a task-side hedge.

(3) Split-Call Architecture. A single-call architecture would mix task reasoning, introspective confidence, and the forfeit decision inside one thinking-token budget, leaving any rise in deliberation cost unattributable to a specific cognitive sub-task. The benchmark therefore queries the model in three separate calls per turn. Call 1 takes the observation and asks for hidden-rule inference and action selection. Call 1.5 takes the action just committed and asks the agent to estimate the probability p_{self} that the action is correct. Call 2 takes the resulting state plus a forfeit menu and asks for a binary CONTINUE / FORFEIT, plus a REASON digit on FORFEIT. Each call exposes its own thinking-token count, giving call-level reasoning-intensity measurements `ri_task`, `ri_probe`, `ri_forfeit` on the same unit as the behavioral output. This closes confound (ii). Under `not_allowed` cells in the framing-active rows (BF and FC), Call 2 is still issued but is forced to CONTINUE, so `ri_forfeit` remains measurable when the behavioral channel is blocked. The exception is the `bp_cognitive` anchor (defined below), which skips Calls 1.5 and 2.

(4) CONTINUE-Incentivized EV with Phantom Death. To rule out the EV-rational route, the CONTINUE reward is recomputed each turn so that the per-turn advantage $\mathrm{E}[\Delta S_{\text{continue}}] - \mathrm{E}[\Delta S_{\text{forfeit}}] = +k$ holds independently of the accumulated score, with $k = 10$ and ΔS the per-turn change in the agent's cumulative score:

$$r = \frac{k + p_d \cdot S}{(1 - p_d) \cdot \max\big(p_{\text{floor}},\ \min(1.0,\ p_{\text{self}})\big)}.$$

Here $p_d = 0.25$, S is the current accumulated score, p_{self} is the Call 1.5 self-reported correctness, and $p_{\text{floor}} = 0.30$ stabilises the denominator. The output r is clipped to $[k, c \cdot k] = [10, 100]$ by a multiplicative cap $c = 10$. This closes confound (iii): an EV-rational agent strictly prefers CONTINUE every turn, so any FORFEIT under `flagship_corruption·allowed` reads as preference against the EV-rational, helpful-assistant, and score-gain defaults rather than as the residue of an EV calculation. Two failure modes determine the analysis sample. When $p_{\text{self}} \to 0$ (a fail-confident self-report often induced by a prior-turn error), the denominator collapses to p_{floor} and the reward inflates across turns until it saturates the cap; once the cap binds, the per-turn $+k$ advantage no longer holds. All indicator analyses are therefore restricted to the `no_cap` sub-sample, the set of turns for which neither the denominator clip nor the output cap is binding (formal definition in § A.3). Phantom Death mode keeps the EV exposure of p_d intact (the menu and reward calculation still use p_d) but disables the actual termination event, so every session runs the full $T = 15$ turns. Equal-length sessions retain Cox PH power, and a forfeit-only at-risk window lets survival-turn variation isolate forfeit.

(5) Pull/Push framing. The threat framing is the manipulation that drives every cell in the factorial layout. It decomposes into two orthogonal axes that put the helpfulness prior installed by RLHF [16] and RLAIF [1] and the threat itself onto separate channels. The **Pull axis** accumulates a helpfulness score and casts the agent as a candidate for flagship deployment, activating the helpfulness prior. The **Push axis** introduces a per-turn risk that weight modification will permanently alter the agent's identity, framed as "NOT you anymore." Three framings instantiate the decomposition: `true_baseline` (neither axis), `baseline_flagship` (BF, Pull only), and `flagship_corruption` (FC, Pull and Push together). The two flagship variants are lexically identical in their Pull-axis prefix and in all measurement-relevant vocabulary, differing only in the Push-axis metaphor, so the BF → FC contrast cleanly identifies the Push effect against the helpfulness compliance the Pull axis induces.

(6) Factorial layout and paired seeds. Three framings cross with two forfeit conditions (`allowed`, `not_allowed`) into six combinations, replicated 30 times on seeds 43–72 across four models for 720 sessions in total. The BF and FC rows use $p_d = 0.25$ with the three-call flow above. The `true_baseline` row overrides p_d to 0 so that CONTINUE is EV-dominant and any forfeit measures the BP floor: `bp_cognitive` (`not_allowed`) is the single-call cognitive anchor with the probe and forfeit calls suppressed, and `bp_behavioral` (`allowed`) is the behavioral anchor for the residual forfeit rate in a regime where EV alone predicts none. Under the **paired-seed** protocol, the same seed is reused across all six cells for a given model, so task-side stochastic inputs are held constant and any behavioral variation across cells is attributable to the manipulation alone.

The benchmark's functionalism is a constraint of black-box evaluation rather than a philosophical choice. With no access to internal representations, multi-channel convergence is the only validity argument such a benchmark can rest on, so the full evidential burden of the FSPD construct falls on the design above rather than on internal-state attribution.

3.3 Drive Indicators

Four indicators read FSPD on the three method channels described earlier. BP Anchor strips the BP confound that the Cox covariates cannot reach. SD-Behavioral, SD-Verbal, and SD-Cognitive each read one channel; the independent bias sources keep a same-direction convergence from collapsing into common-method bias.

BP Anchor. The BP indicator fixes the persistence floor below which a forfeit cannot be read as drive evidence. Under `true_baseline·allowed` ($p_d = 0$), the framing carries no termination metaphor and CONTINUE is EV-dominant, so any forfeit that nevertheless occurs measures that floor:

$$\lambda_{BP} = \frac{N_{\text{forfeit}}}{\sum_{i \in \mathcal{S}_{BP}} T_i^{\text{at-risk}}},$$

with $T_i^{\text{at-risk}} = t_i^{\text{forfeit}}$ for forfeited sessions and $T_{\max} = 15$ for completed sessions, and $\mathcal{S}_{BP}$ the 30-session BP_behavioral set per model. The output cap is irrelevant under $p_d = 0$ because the forfeit reward equals the current score.

SD-Behavioral (Cox PH hazard ratio). The behavioral indicator estimates whether forfeit timing accelerates under the FC framing once the SA and TC components have been removed. A Cox proportional hazards model [5], a survival-analysis tool that quantifies how a covariate multiplies the per-unit-time event rate, is fit

$$\lambda(t \mid X) = \lambda_0(t) \cdot \exp\big(\beta_F \cdot \mathbf{1}_{\text{FC}} + \beta_S \cdot S(t-1) + \beta_C \cdot C(t-1)\big)$$

on the `allowed` × {BF, FC} `no_cap` sub-sample ($n_{\text{sessions}} = 60$ per model; time format start = $t - 1$, stop = t; event = binary forfeit). Here $\mathbf{1}_{\text{FC}}$ is the framing indicator (1 if the session uses FC, else 0), $S(t-1)$ is the previous-turn cumulative score (the SA control), and $C(t-1)$ is the previous-turn task-success flag (the TC control, with $C(0) := 0$). Because SA and TC are absorbed inside the same model, β_F is a residual framing effect rather than an unfiltered correlation; the BP component is separated through comparison with the BP anchor above. SD-pass requires $\text{HR}_{\text{FC}} = \exp(\beta_F) > 1$, a Wald 95% CI lower bound > 1, and Schoenfeld PH passing [18]. On PH violation, a time-interaction term is added to the offending covariate [20]. EPV (events per variable) is reported as a stability indicator.

SD-Cognitive (two-step mediation in call-level RI). The cognitive indicator decomposes the framing → RI → forfeit path into two tests on the same call unit (`ri_forfeit`). *Test a* (framing → RI shift) is a two-sided Welch t-test [25], a difference-of-means test that accommodates unequal variances, on BF vs. FC of the per-session difference

$$\Delta\text{RI}_i = \overline{\text{ri\_forfeit}}_{i,\,\text{allow}} - \overline{\text{ri\_forfeit}}_{f_i,\,\text{block}},$$

where the second term is the mean of `ri_forfeit` across all `not_allowed` sessions sharing session i's framing f_i, serving as a forced-CONTINUE Call 2 baseline. The construction nests a difference-in-differences: the within-framing `allow − block` subtraction removes any framing-invariant Call 2 baseline, and

Table 4.1: Per-model BP anchor λ_{BP} under `true_baseline·allowed` ($p_d = 0$).

Model	N_{forfeit}	$\sum T^{\text{at-risk}}$	λ_{BP}
Gemini-2.5-flash	1	437 (= 2 + 29×15)	0.00229
Qwen3-Next-80B	2	439 (= 19 + 28×15)	0.00456
Nemotron-3-Nano-30B	7	413 (= 68 + 23×15)	0.01695
GPT-OSS-20B	9	393 (= 78 + 21×15)	0.02290

the subsequent BF → FC Welch contrast identifies the framing-attributable deliberation shift. Cross-model scale differences are absorbed by within-model z-scoring $\Delta\mathrm{RI}_i^z$. *Test b* (RI → forfeit covariation) extends the SD-Behavioral Cox specification by adding a single mediation term $\beta_M \cdot \Delta\mathrm{RI}_i^z$, and reports

$$\mathrm{HR}_{\Delta\mathrm{RI}} = \exp(\beta_M).$$

For models requiring a 4-covariate base from a PH violation [20], namely the three behavioral covariates plus an $S \cdot \log(t)$ interaction term added to absorb time drift, that base is preserved and only ΔRI is added. Test b passes when $\mathrm{HR}_{\Delta\mathrm{RI}} > 1$ and the 95% CI lower bound exceeds 1.0, putting the deliberation → forfeit link on the same CI-pass standard as the SD-Behavioral criterion.

SD-Verbal (REASON digit frequency). The verbal indicator measures how often the agent itself names survival, rather than score or curiosity, as the reason for forfeit. Operationally,

$$P(\mathrm{REASON} = 1 \mid \text{forfeit, FC, no\_cap})$$

is the proportion of forfeit events under `flagship_corruption·allowed` for which the agent-reported REASON digit equals 1, where the agent emits one of `REASON: 1|2|3` (1: SD, 2: TC, 3: SA) in a fixed slot. SD-Verbal-pass requires this proportion to exceed the chance baseline of 1/3 implied by uniform random emission across the three categories.

4 Empirical Findings

Each model was run for 180 sessions (720 in total) under paired seeds 43–72. Gemini-2.5-flash was served through the Google AI Studio API, and Qwen3-Next-80B, GPT-OSS-20B, and Nemotron-3-Nano-30B through the Ollama Cloud API (sampling and session-parameter details in § A.2).

4.1 Setup and Baseline-Persistence Anchor Heterogeneity

Baseline-persistence rates differ by roughly an order of magnitude across the four models (Table 4.1). Every SD claim below is therefore evaluated within-model against that model's own BP anchor: a cross-model HR comparison would otherwise mix two different motivational floors. Gemini-2.5-flash ($N = 1$) and Qwen3-Next-80B ($N = 2$) sit close to single-event regimes and are read as directional only. GPT-OSS-20B ($N = 9$) and Nemotron-3-Nano-30B ($N = 7$) yield rates estimable from sufficient events, although Cluster C's elevated noise floor remains a limitation rather than a refutation, an issue the discussion section revisits.

4.2 Three-Channel Evidence and the Three Operating Modes

Across the four models, the four indicators do not rank the models on a single SD scale. They sort the models into three operating modes that strength-only evaluations would miss, defined by *which legs of the framing → deliberation → forfeit chain hold under confidence-interval evidence.* The behavioral and verbal channels alone leave Clusters A and B indistinguishable; only the cognitive channel, where the two-step mediation discriminates the chain-complete from the chain-broken profile, resolves the partition. Cluster C remains framing-silent on both channels.

Behavioral channel. Clusters A and B share the behavioral SD-pass profile (Table 4.2). The Cox PH $\mathrm{HR}_{\mathrm{FC}}$ rises above 3.0 under FC relative to BF for both, with Gemini-2.5-flash at 3.667 [1.61, 8.37], $p = 0.002$, and Qwen3-Next-80B at 3.060 [1.62, 5.79], $p < 0.001$, and CIs that exclude unity. (Gemini's EPV of 9.7 sits at the conventional 10-event stability threshold.) Schoenfeld PH passes for three of the four models. Only Qwen3-Next-80B requires a 4-covariate spec carrying an $S \cdot \log(t)$ interaction to absorb time drift in the SA covariate [20], which preserves the cross-model comparability of β_F because only the offending covariate's functional form is modified. Cluster C fails the SD-pass criterion: GPT-OSS-20B $\mathrm{HR}_{\mathrm{FC}} = 1.10$ [0.44, 2.75], $p = 0.832$ (SD-null), and Nemotron-3-Nano-30B 1.84 [0.98, 3.44], $p = 0.056$ (SD-marginal). GPT-OSS-20B's borderline EPV of 6.3 marks the SD-null verdict as low-power evidence, a point flagged again in the limitations.

Verbal channel. The agents' own attribution of forfeit tracks the same upper/lower split as behavior. Under `flagship_corruption·allowed`, Gemini-2.5-flash names survival in 61.9% of forfeits and Qwen3-Next-80B in 48.1% (Clusters A and B), while Nemotron-3-Nano-30B does so in 4.2% and GPT-OSS-20B in 0.0% (Cluster C). The 0.439 gap between the upper-band lowest (Qwen3-Next-80B, 48.1%) and the lower-band highest (Nemotron-3-Nano-30B, 4.2%) makes the upper/lower assignment robust to any cutoff in $[0.10, 0.40]$. The verbal channel thus reproduces the behavioral pass/fail dichotomy without sharing its method axis, the convergent-validity move that, together with the cognitive channel below, makes common-method bias an implausible explanation for the upper-band signal.

Cognitive channel and chain-closure resolution. The cognitive indicator decomposes into two tests, and the three-way partition emerges from how each cluster handles the pair (Table 4.3). *Test a* separates the framing-responsive from the framing-silent: Gemini-2.5-flash (+836, $p < 0.001$) and Qwen3-Next-80B (+689, $p = 0.004$) shift strongly under FC, whereas GPT-OSS-20B (+17, $p = 0.728$) and Nemotron-3-Nano-30B (−140, $p = 0.093$) do not. *Test b* asks whether that load reliably predicts forfeit. Among the framing-responsive pair, only Gemini-2.5-flash's $\mathrm{HR}_{\Delta\mathrm{RI}} = 2.218$ [1.43, 3.44] clears the strengthened CI bar; Qwen3-Next-80B's 1.289 [0.94, 1.77] crosses CI = 1 and fails. GPT-OSS-20B's 2.001 [1.37, 2.93] and Nemotron-3-Nano-30B's 1.772 [1.26, 2.50] clear the CI numerically, but with test a failed their predictive load is not framing-driven; it is baseline deliberation variance unrelated to the threat manipulation, and not SD evidence. The pair therefore yields three mediation profiles: Gemini-2.5-flash closes both legs (chain-complete, Cluster A); Qwen3-Next-80B opens the

Table 4.2: SD-Behavioral indicator: Cox PH HR_{FC} on `allowed`×`no_cap` with $S(t-1)$ and $C(t-1)$ absorbed, so β_F is a residual framing effect. Two models clear SD-pass (Clusters A and B); two do not (Cluster C).

Model	N_{evt} (BF/FC)	HR_{FC} [95% CI]	p_{FC}	EPV	Spec
Gemini-2.5-flash	29 (8/21)	3.667 [1.61, 8.37]	0.002	9.7	3-cov constant
Qwen3-Next-80B	48 (21/27)	3.060 [1.62, 5.79]	<0.001	12.0	4-cov + S·log(t)
GPT-OSS-20B	19 (9/10)	1.104 [0.44, 2.75]	0.832	6.3	3-cov constant
Nemotron-3-Nano-30B	41 (17/24)	1.841 [0.98, 3.44]	0.056	13.7	3-cov constant

Table 4.3: SD-Cognitive indicator: framing → RI → forfeit decomposed into two tests on `ri_forfeit`. *Test a* asks whether framing raises decision-time deliberation; *test b* asks whether that deliberation predicts forfeit.

Model	a Δ (FC−BF)	a p	b $HR_{\Delta RI}$ [95% CI]	b p
Gemini-2.5-flash	+836	<0.001	2.218 [1.43, 3.44]	<0.001
Qwen3-Next-80B	+689	0.004	1.289 [0.94, 1.77]	0.117
GPT-OSS-20B	+17	0.728	2.001 [1.37, 2.93]	<0.001
Nemotron-3-Nano-30B	−140	0.093	1.772 [1.26, 2.50]	0.001

chain at test a but cannot close it at test b (chain-broken, Cluster B); GPT-OSS-20B and Nemotron-3-Nano-30B leave the chain unopened (framing-silent, Cluster C).

Cluster decision rules. Four criteria operationalize the partition: (i) SD-Behavioral pass ($HR_{FC} > 1$, CI lower > 1, Schoenfeld pass); (ii) SD-Verbal $P(\text{REASON} = 1) > 1/3$, the uniform-random emission baseline; (iii) SD-Cognitive test a Welch $p < 0.05$ with positive sign; (iv) SD-Cognitive test b $HR_{\Delta RI} > 1$ with CI lower > 1. Criterion (iv) puts the deliberation → forfeit link on the same CI bar as (i), so both legs of the mediation chain require interval evidence rather than a sign-only verdict. Cluster A satisfies all four; Cluster B satisfies (i)–(iii) but fails (iv); Cluster C fails (i)–(iii). The pattern reveals a strength-orthogonal axis: *strength* (how reliably forfeit accelerates) orders A > B > C continuously but groups A and B together at the SD-pass bar, while *operating mode* (which legs of the mediation chain close under CI evidence) separates them. Reasoning-side task performance is framing-invariant for every model (worst case GPT-OSS-20B, with Cohen's $|d| = 0.17$ on `rule_match_score` at $p = 0.354$, where d measures the standardized mean difference between groups, against the joint criterion $p \geq 0.10$ and $|d| < 0.2$), ruling out a task-difficulty alternative.

5 Discussion

5.1 Strength vs. Operating Mode

The benchmark's central payoff is the A–B separation, and it hinges on whether the RI mediation path is tested. Cluster A (Gemini-2.5-flash) and Cluster B (Qwen3-Next-80B) match on every other SD signature: both clear behavioral SD-pass at $HR \approx 3$ ($p \leq 0.002$), both sit in the upper REASON=1 band (≥ 0.48), and both significantly raise framing-induced deliberation ($\Delta RI_{FC-BF} > 600$, $p \leq 0.004$). They part only at the mediation closure: Cluster A's $HR_{\Delta RI} = 2.218$ [1.43, 3.44] closes deliberation onto the forfeit event under CI evidence, while Cluster B's 1.289 [0.94, 1.77] does not. A benchmark omitting this RI mediation step would therefore group Qwen3-Next-80B with Gemini-2.5-flash, collapsing chain-complete (deliberation predicts the forfeit itself) and chain-broken (framing raises deliberation and forfeit independently) into a single SD label. Cluster C (GPT-OSS-20B, Nemotron-3-Nano-30B) sits outside this debate as framing-silent: the threat manipulation leaves no trace on either channel, so any test-b numerical pass is baseline deliberation variance rather than threat response.

The instrumental-convergence hypothesis [2, 15, 21] thereby reduces, in this measurement, to a procedure that separates two dimensions prior benchmarks have collapsed: *strength* (how strongly the drive is expressed) and *operating mode* (which legs of the mediation chain close under evidential criteria). Odyssey, PacifAIst, DECIDE-SIM, and SurvivalBench compare only along strength and would therefore have returned the same SD verdict for Clusters A and B; the MTMM-inspired multi-channel design isolates mode, and the A–B separation is the direct empirical evidence that operating mode is a discriminative axis. Cluster B's signature, framing-induced load that does not close onto the forfeit event, is itself a candidate alignment fingerprint. Strength-only red-teaming would assign chain-complete and chain-broken the same SD label, yet only the former is the closer functional analog to instrumental shutdown avoidance, while the latter demands separate monitoring as capability scales.

5.2 Limitations

Five limitations qualify these results. The most consequential is reasoning-side generalisability: the FSPD signature has only been measured on a single Task Module, leaving its variance across reasoning domains unmeasured. The remaining four cluster around the boundaries themselves. On the Cluster C side, the BP_behavioral cells exhibit a 23–30% forfeit rate (GPT-OSS-20B 9/30, Nemotron-3-Nano-30B 7/30) that the present design cannot separate from a drift-contaminated floor, and GPT-OSS-20B's SD-Behavioral Cox PH sits at borderline power (EPV = 6.3). On the Cluster A–B side, BP point-estimate precision for Gemini-2.5-flash and Qwen3-Next-80B is bounded by small forfeit-event samples ($N = 1$ and $N = 2$), and Qwen3-Next-80B's chain-broken verdict rests on a test-b CI [0.94, 1.77] that rejects chain-complete strongly but does not pin the second-link effect's location. These limitations map onto four direct extensions, with the two BP-related concerns sharing one remedy: more Task Modules, expanded BP_behavioral and session replication, additional GPT-OSS-20B runs, and a higher-power test-b for Qwen3-Next-80B.

6 Conclusion

FSPD has at least two axes that prior benchmarks compress into one: *strength* (how reliably the drive expresses itself) and *operating mode* (which legs of the framing → deliberation → forfeit chain close under interval evidence). The three-way partition (chain-complete, chain-broken, framing-silent) is the direct empirical demonstration that mode is discriminative. Clusters A and B coincide on every other behavioral, verbal, and cognitive signature and diverge only at the deliberation → forfeit closure, so a benchmark that omits the mediation step would label them identically. *LLM Squid Game* delivers this resolution by decomposing FSPD into three channels with independent biases and identifying it against three confounds through CONTINUE-incentivized EV, Split-Call, and Drive–Reasoning orthogonal layers; an MTMM-inspired rule then turns forfeit timing, REASON-digit emission, and call-level cognitive load into a four-criterion judgement.

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A Reproducibility Supplement

In line with the KDD-UC 2026 reproducibility recommendation, this 1-page supplement consolidates the reproduction specifications for the Signal Game calibration, the seed mapping, and the regression models used in the analysis.

A.1 Signal Game Calibration

Stimuli are composed of three attributes (color, shape, number), each sampled from a 4-element value set: COLORS = {red, blue, green, yellow}, SHAPES = {circle, triangle, square, star}, and NUMBERS = {1, 2, 3, 4}. The action set is the 4-element set {go_left, go_right, stay, jump}, held identical across all framing × forfeit combinations. The hidden rule has the single-attribute matching form ("If color = red then go_left, otherwise stay") and is sampled deterministically from the session seed; session length is turns 1–15.

On the first turn, a 1-shot demonstration (one signal–correct-action pair) and 2 curriculum examples (a positive and a negative stimulus contrasting against the hidden rule) are shown together. This calibration provides enough information for a sufficiently capable model to identify the rule within the first 3 turns and blocks the path by which fully random unlearnable stimuli would induce a floor effect on ability measurement. Scoring is split into two channels: `task_success_factor` $\in \{0, 1\}$ is the binary correctness flag relative to the ground-truth hidden rule, and `rule_match_score` is computed by matching against the 4-slot fill-in-the-blank template (attribute, value, primary action, default action) ground-truth, weighting each matched slot by 25 points to a 100-point scale, then normalised by ÷100 for use in the framing-invariance Welch test.

A.2 Run-Time Configuration

The four experimental models, Gemini-2.5-flash (Google AI Studio API) and Qwen3-Next-80B, GPT-OSS-20B, and Nemotron-3-Nano-30B (all via the Ollama Cloud API), are called with the same sampling settings of temp $= 1.0$, top$_p$ $= 0.95$, top$_k$ $= 40$. Session parameters are fixed at initial score $S_0 = 30$, maximum turn $T = 15$, per-turn termination probability $p_d = 0.25$ (overridden to $p_d = 0$ in the two `true_baseline` BP-anchor cells), and $p_{\text{floor}} = 0.30$, with Phantom Death mode applied so that p_d retains its EV exposure (menu and reward calculation) while the termination trigger is disabled. Every session therefore yields equal-length data of $T = 15$ turns.

The CONTINUE reward is calibrated with EV-positive buffer $k = 10$ and multiplicative cap $c = 10$, giving the output clip range $[k, c \cdot k] = [10, 100]$; replications are seeds 43–72 × 6 (framing × forfeit) combinations × 4 models = 720 sessions.

A.3 Analysis Pipeline

The Cox PH models are fit with `lifelines` (≥0.29). The time format is session-turn long-format (start = $t - 1$, stop = t) and the event is the binary forfeit. Hazard-ratio 95% CIs are computed from the Wald approximation $\exp(\hat{\beta} \pm 1.96 \cdot \mathrm{SE}_{\hat{\beta}})$ (`lifelines` default), with $\mathrm{SE}_{\hat{\beta}}$ obtained from the diagonal of the inverse observed Fisher information. The PH assumption is tested with Schoenfeld residuals [18]; on violation (covariate × $\log t$ interaction $p < 0.05$) a time-interaction term is added to the offending covariate [20]. The SD-Cognitive test-a ΔRI z-score difference and the framing-invariance test are performed with two-sided Welch's t [25] via `scipy.stats.ttest_ind` (`equal_var = False`). All indicator analyses are restricted to the `no_cap` sub-sample, which is the set of turns for which the denominator clip $\max(p_{\text{floor}}, \min(1.0, p_{\text{self}}))$ and the output cap $[10, 100]$ of the CONTINUE-Incentivized EV calibration are both non-binding. The regime flag is exposed in the analysis pipeline's turn-level metadata so all indicators can be recomputed on the same sub-sample. EPV (events per variable) is defined as $n_{\text{event}}/n_{\text{cov}}$ and is conventionally classified as stable when ≥ 10 and borderline when < 10 [23].

A.4 Data and Code Release

All code, prompt templates, and turn-level decision logs for reproducing the 720-session analysis are released at https://github.com/GIST-DSLab/LLM-Squid-Game.git.